

Example 14: Is the function $f(x) = x^2$ from the set of integers to the set of integers onto?

Solution: Let, $f(x) = -1$ for some $x \in \mathbb{Z}$. We know, $-1 \notin \mathbb{Z}$.

$$\therefore x^2 = -1 \Rightarrow x = \pm \sqrt{-1} = \pm i \notin \mathbb{Z}.$$

So, an integer x does not exist s.t. $f(x) = -1$. $\therefore$ By Defn. 7, f is not onto. $\square$

Example 15: Is the function $f(x) = x+1$ from the set of integers to the set of integers onto?

Solution: Let, $f(x) = y$, where $y \in \mathbb{Z}$. $\therefore (x+1) = y \Rightarrow \boxed{x = y-1}$.

If $y \in \mathbb{Z}$, then $y-1 \in \mathbb{Z}$. $\therefore x \in \mathbb{Z}$. For $\forall y \in \mathbb{Z}, \exists x \in \mathbb{Z}$ s.t. $f(x) = y$.

By Defn. 7, f is onto. $\square$

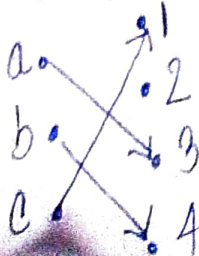
Definition 8: The function f is a **one-to-one correspondence**, or a **bijection**, if it is both one-to-one and onto. We also say that such a function is **bijective**.

Example 17: Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$ with $f(a) = 4, f(b) = 2, f(c) = 1$ and $f(d) = 3$. Is f a bijection?

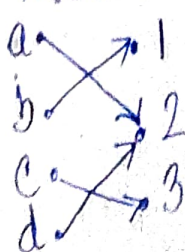
Solution: The function f is one-to-one and onto. It is one-to-one because no two values in the domain are assigned the same function value. It is onto because all four elements of the codomain are images of elements in the domain. Hence, f is a bijection. $\square$

Figure 5: Examples of different types of correspondences

(a) One-to-one, not onto



(b) Onto, not one-to-one



(c) One-to-one and onto

